

AMENDMENT NO. 2 APRIL 2023
TO
IS 15140 : 2018 AUTOMOTIVE VEHICLES — SAFETY BELT AND RESTRAINT
SYSTEMS — SPECIFICATION
(First Revision)

[Page 9, clause **4.4.2.9** (c)] — Insert the following new clause at the end of the clause:

‘d) In case of a seating position, other than the outboard front seating position, if fitted with an airbag in front of it, the displacement of the chest reference point may exceed that specified in **4.4.2.9** (b), if its speed at this value does not exceed 24 km/h in tests performed with the airbag in a vehicle related environment, reflecting the vehicle coordinates of the airbag mounting and attachment points.’

(Page 9, clause **4.4.2.10**) — Substitute the following for the existing clause:

‘4.4.2.10 In the case of a restraint system

The movement of the torso reference point may exceed that specified in **4.4.2.9** (b), if it can be shown either by calculation or a further test that no part of the torso or the head of the manikin used in the dynamic test would have come into contact with:

- a) In the case of the driver any forward rigid part of the vehicle other than the chest with the steering assembly, if the latter meets the requirements of IS 11939 and, provided, the contact does not occur at a speed higher than 24 km/h. For this assessment the seat shall be considered to be in the position specified in **5.5.2.6**.
- b) In the case of any other occupant, contact of the head or of the chest with any rigid part of the vehicle in front of the manikin is not allowed. Additionally, contact of the manikin head with its knees is not allowed. For this assessment the seat of the tested manikin, and, if applicable, the seat in front of the manikin shall be considered to be in the positions specified in **5.5.2.6**, with the exception of the deployed structure of an airbag assembly defined in **3.24**, non-rigid material of < 50 Shore A hardness may be removed in order to demonstrate that covered or padded rigid parts are not contacted by the head or the chest of the manikin during the test.
- c) In vehicles where such devices are used, the displacement and locking system enabling the occupants of all seats to leave the vehicle shall still be operable by hand after the dynamic test.’

(Page 20, clause **5.5.2.6**) — Substitute the following for the existing clause:

‘5.5.2.6 For the assessment of the requirements:

- a) In **4.4.2.10** (a), the seat shall be regarded as being in its most forward driving or travelling position appropriate to the dimensions of the manikin.
- b) In **4.4.2.10** (b), the seat adjustment for front passenger seats shall be as follows:
The occupant in the front shall be in its most forward travelling position according to the dimensions of the manikin. The positions of the seats shall be stated in the report.
- c) In **4.4.2.10** (b), the seat adjustment for rear seats shall be as follows:
For any testing position of rear seated occupants, the position of the tested seating position, shall be considered as positioned to the seats R-Point. If the seat back is adjustable, the seat back angle should be adjusted using the 3-D H Point machine with its torso angle as close as possible to 10°.

The seating position in front of the tested seating position shall be adjusted to the middle position of travel and height or in the nearest locking position thereto. If the seat back of this seat is adjustable, the seat back angle should be adjusted using the 3-D H Point machine with its torso angle as close as possible to 10°. This may be proven by a contour on CAD or drawing of the in-vehicle situation.’